Problems

Section 15 — Batteries

Problem 15.1

A battery with a capacity of 1800 mAh is to be used in a device that draws 120 mA continuously. Ignoring possible loss in capacity as a result of load current magnitude, how long should the battery be able to deliver power?

Problem 15.2

Determine the discharge time and average current output of a battery with a capacity rating of 1000 mAh if it is discharged at 1 C.

How long would it take to discharge at 5 C, 2 C, 0.5 C, 0.2 C and 0.05 C?

Section 16 — Resistors

Problem 16.1

If a 1000 Ω resistor is connected in parallel with one of 3000 Ω, what is the total (equivalent) resistance?

Problem 16.2

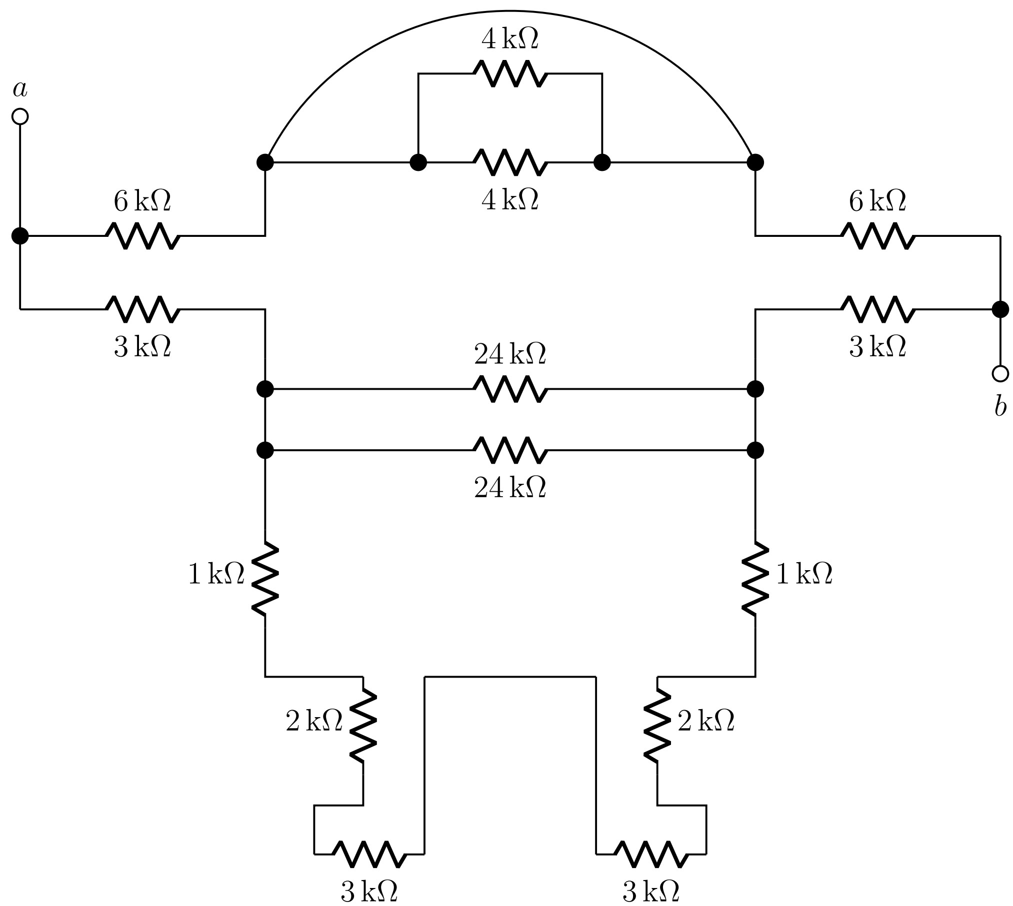
If a 1.0 kΩ resistor is placed in series with a 2.0 kΩ resistor, what is the total resistance?

If the two resistors are then attached to an input voltage of $V_{in} = 9$ V, find:

1. The total current flow and the current through each resistor.
2. The voltage drop across each resistor.

Problem 16.3

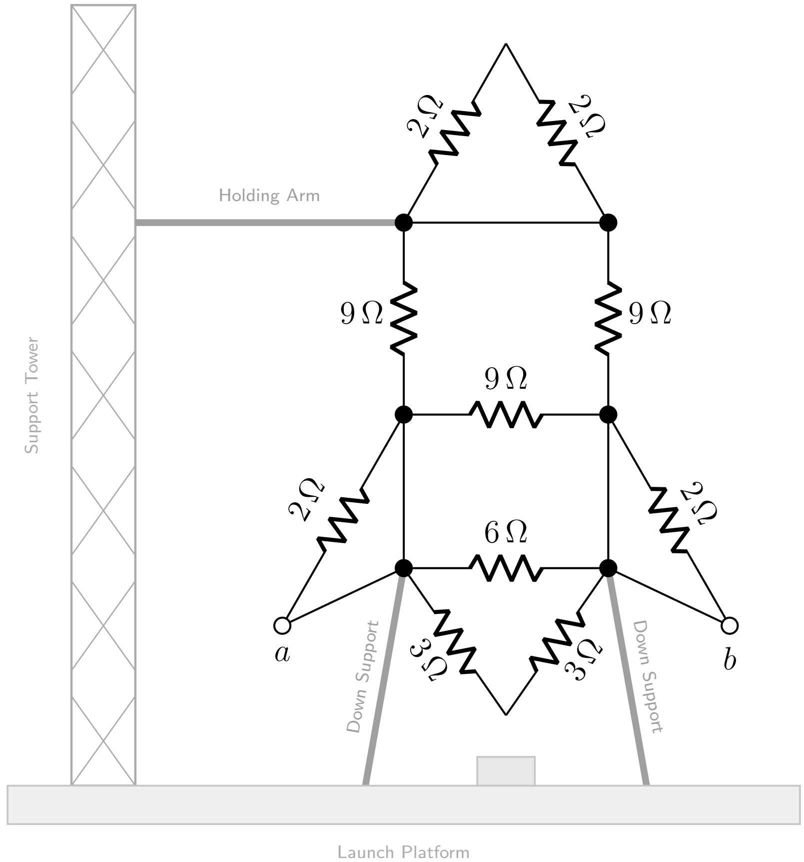
Find the equivalent resistance between terminals a and b of the resistor network shown below.



Problem 16.4

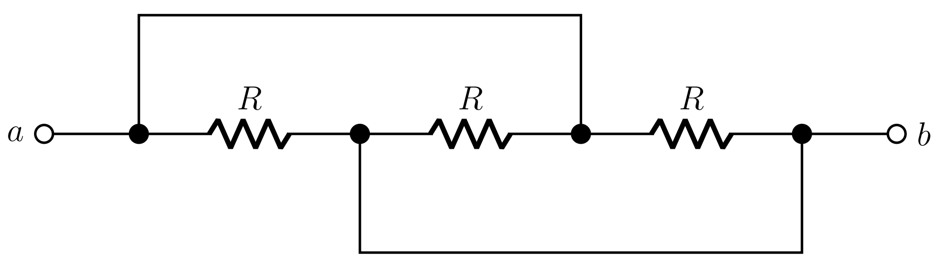
Find the equivalent resistance between terminals a and b of the resistor network shown below. All resistances are given in ohms (Ω).

The support tower, holding arm, down supports and launch platform (drawn in grey) are mechanical structures only — they are **not** electrical connections.



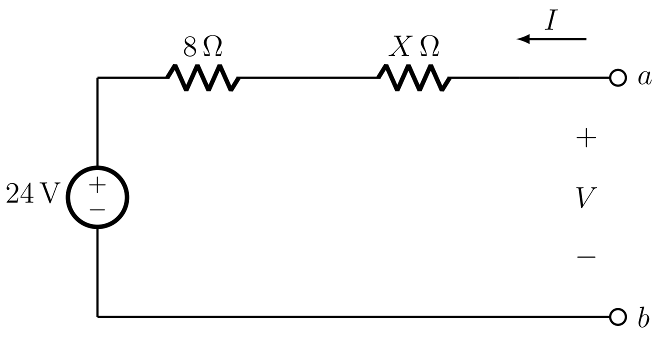
Problem 16.5

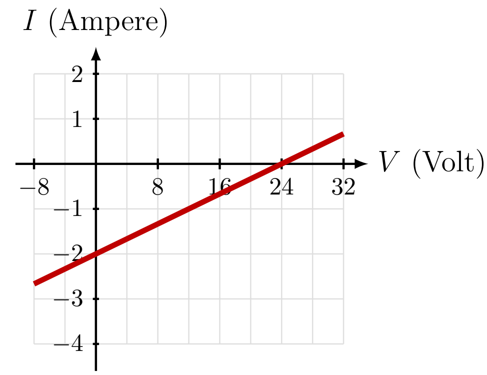
In the circuit below, all three resistors have the same resistance R. Find the equivalent resistance between the terminals a and b. What is the configuration of the three resistors — series, parallel, or neither?



Problem 16.6

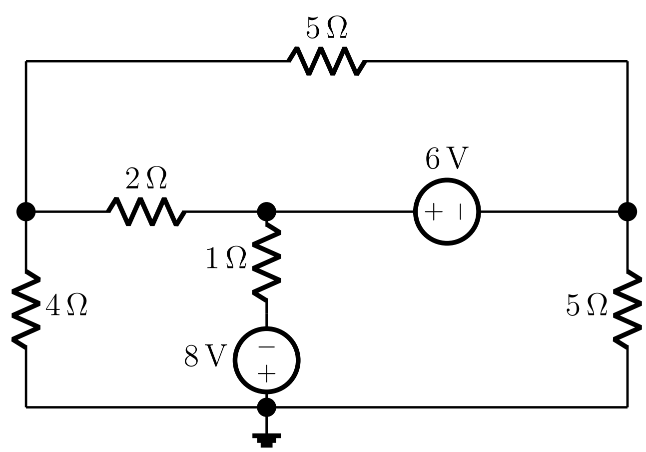
The two-terminal network below consists of a 24 V source in series with an 8 Ω resistor and an unknown resistance X. The terminal current I and the terminal voltage V, measured at terminals a–b with the polarities marked in the figure, obey the straight-line relationship shown alongside. Determine the value of the unknown resistance X.





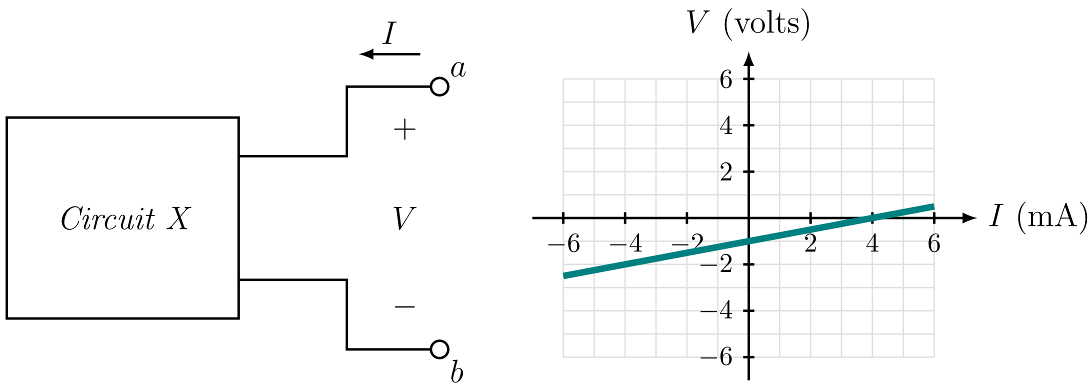
Problem 16.7

Apply nodal (or mesh) analysis to the circuit below to determine all of the node voltages — or, if you choose to work with meshes, all of the mesh currents. The bottom rail, marked with the ground symbol, is the reference node.

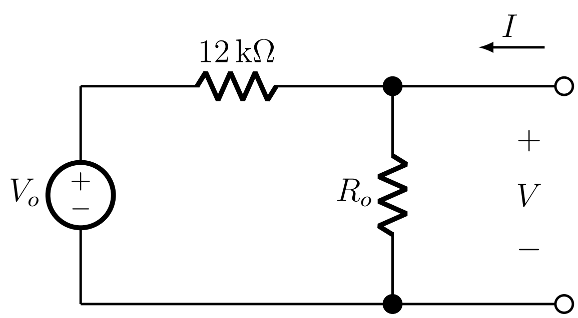


Problem 16.8

Circuit X is a linear two-terminal circuit connected between terminals a and b, as shown below. When tested, the circuit follows the voltage (V) and current (I) relationship shown in the accompanying plot.

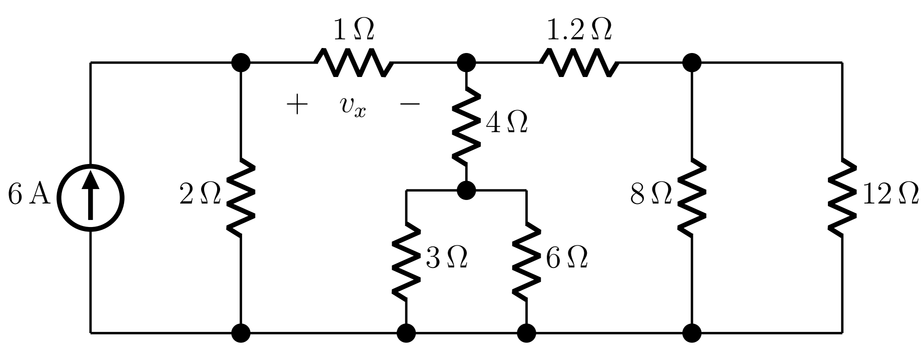


1. Identify the range of the voltage V for which the circuit behaves as an **active** circuit (i.e. supplying power).
2. Analyse the network below to determine the values of R_o and V_o, assuming this network is equivalent to *Circuit X*.



Problem 16.9

Find the voltage v_x across the 1 Ω resistor in the circuit below, with the polarity marked (+ on the left). The source supplies a constant 6 A.

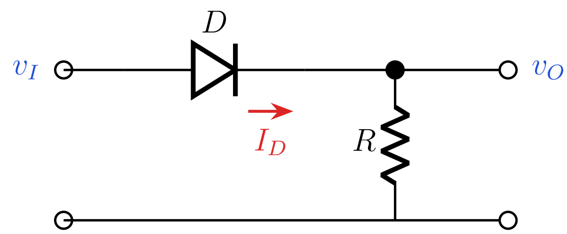


Section 20 — Diodes

In every problem, validate your assumptions after solving.

Problem 20.1

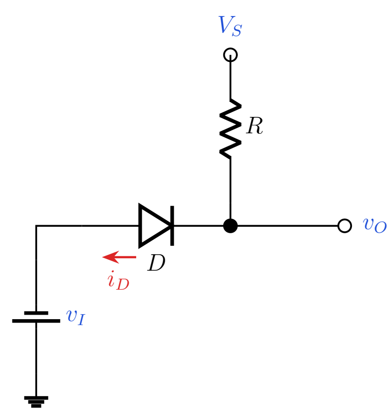
For the circuit below, if $v_I = 5$ V and $V_{DO} = 0.7$ V, find v_O.



Problem 20.2

For the circuit below, find v_O in each of the following cases. Take $V_{DO} = 0.7$ V throughout.

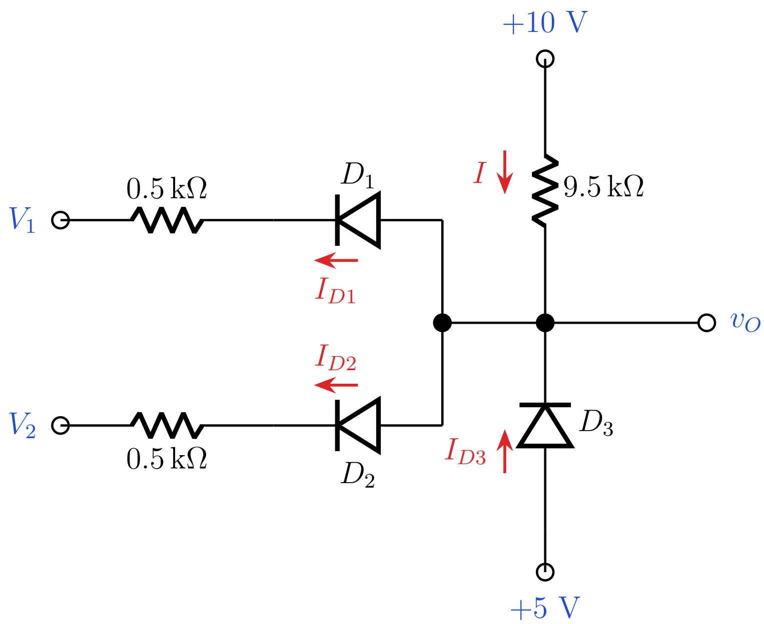
1. $v_I = 5$ V, $V_S = 10$ V
2. $v_I = 11$ V, $V_S = 10$ V
3. $v_I = 0$ V, $V_S = 10$ V



Problem 20.3

Analyse the circuit below to find I_{D1}, I_{D2}, I_{D3} and I for each of the following input combinations. Use the **CVD+R model** with $V_{DO} = 0.6$ V and $r_d = 50\ \Omega$.

1. $V_1 = 0$ V, $V_2 = 2$ V
2. $V_1 = V_2 = 0$ V
3. $V_1 = V_2 = 5$ V
4. $V_1 = 5$ V, $V_2 = 0$ V



Problem 20.4

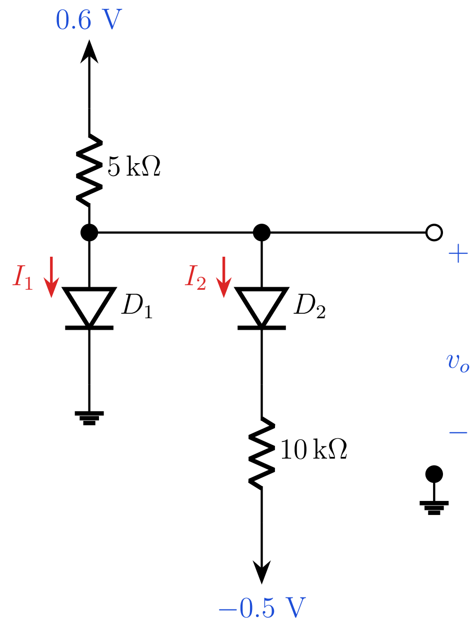
For the same circuit as Problem 20.3, replace the 9.5 kΩ pull-up resistor with a resistor of value X (in kΩ) and take $V_{DO} = 0.6$ V.

1. Repeat the analysis of Problem 20.3 for $X = 9.5$.
2. For what value of X will D_3 be in cut-off?

Problem 20.5

Analyse the circuit below to find I_1, I_2 and v_o under each of the following diode models.

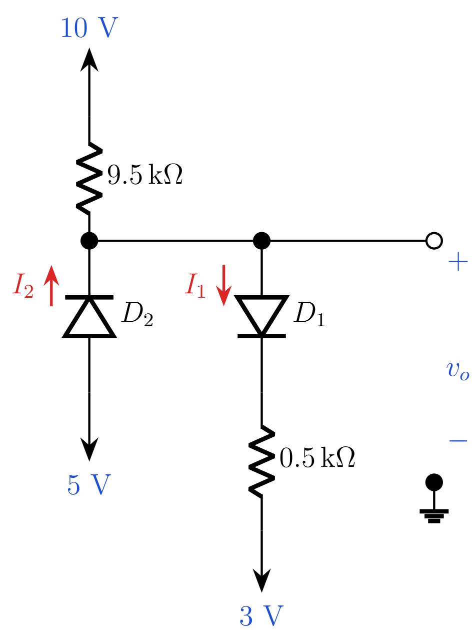
1. All diodes assumed **ideal**.
2. **CVD model** with $V_{D1} = 0.5$ V and $V_{D2} = 0.7$ V.
3. **CVD+R model** with $V_{DO} = 0.6$ V and $r_d = 50\ \Omega$.



Problem 20.6

Analyse the circuit below to find I_1, I_2 and v_o under each of the following diode models.

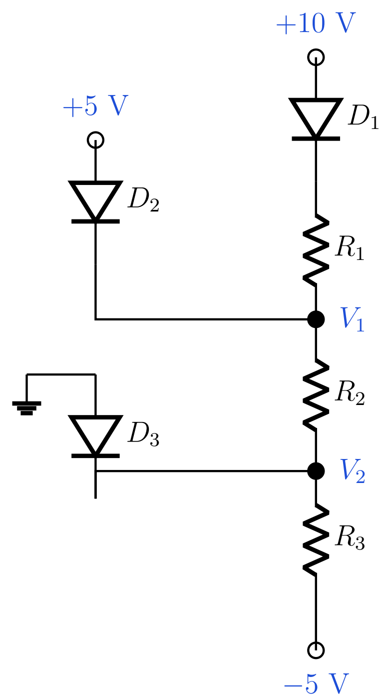
1. All diodes assumed **ideal**.
2. **CVD model** with $V_{DO} = 0.6$ V.
3. **CVD+R model** with $V_{DO} = 0.6$ V and $r_d = 50\ \Omega$.



Problem 20.7

For the circuit below, take $V_{DO} = 0.6$ V.

1. Determine R_1, R_2 and R_3 such that $I_{D1} = 0.2$ mA, $I_{D2} = 0.3$ mA and $I_{D3} = 0.5$ mA.
2. Find V_1, V_2 and each diode current for $R_1 = 10$ kΩ, $R_2 = 4$ kΩ and $R_3 = 2.2$ kΩ.



Problem 20.8

For the same circuit as Problem 20.7 with $V_{DO} = 0.6$ V, find V_1, V_2 and each diode current for:

1. $R_1 = 3\ \text{k}\Omega,\ \ R_2 = 6\ \text{k}\Omega,\ \ R_3 = 2.5\ \text{k}\Omega$
2. $R_1 = 6\ \text{k}\Omega,\ \ R_2 = 3\ \text{k}\Omega,\ \ R_3 = 6\ \text{k}\Omega$

Section 21 — Rectifiers

Validate your assumptions where a diode state must be chosen.

Problem 21.1

A voltage waveform $v_i = 10\sin(100\pi t)$ V is input to a full-wave rectifier with a load resistance of $R = 50$ kΩ. Silicon diodes are used in this circuit, for which the forward drop is $V_{D_0} = 0.7$ V.

1. **Show** the circuit of the rectifier. **Label** the input and output voltages properly. [2]
2. **Calculate** the DC value of the output voltage. [1]
3. **Contrast** the value found in part (b) with that when a 5 μF capacitor is connected in parallel with the load. [2]

4. **Identify** the two diodes that will be ON in the positive half cycle. [1]

Now the two diodes from part (d) are replaced with germanium diodes ($V_{D_0} = 0.2$ V).

5. **Explain** the change in the voltage transfer characteristics and output voltage waveform of the circuit. Hence, calculate the peak of the output voltage in this case. [3+1]

Problem 21.2

A voltage waveform $v_i = 5\sin(200\pi t)$ V is fed into a full-wave rectifier with a load resistor $R = 5$ kΩ. Silicon diodes are used in this circuit, where $V_{D_0} = 0.6$ V.

1. **Draw** the rectifier circuit. **Label** the input and output voltages properly. Briefly **explain** the application of the circuit. [1+1+1]
2. **Calculate** the DC value of the output voltage, V_{dc}, and the output frequency, f_o. [1+1]
3. **Draw** the Voltage Transfer Characteristics (VTC) of the full-wave rectifier and **label** it properly. [2]
4. Now you have to connect a capacitor in parallel with the load resistor. You have two capacitors of 5 μF and 1 μF at your disposal. Which capacitor will you use? **Explain** briefly with necessary calculations. [3]
5. [**Bonus**] A different input waveform is fed into the full-wave rectifier. The new peak-to-peak ripple voltage is 50% of the previous one calculated in part (d) with the 5 μF capacitor. The new output frequency is 300 Hz. **Determine** the equation of the input waveform. [2]

Problem 21.3

A voltage waveform $v_i = 10\sin(1000\pi t)$ V is fed into a full-wave (FW) rectifier with a load resistance $R = 10$ kΩ. A capacitor is also connected in parallel with the load to reduce the fluctuation of the output voltage. It produces a peak-to-peak ripple voltage which is **3**% of the peak output voltage. The diodes have a forward voltage drop of $V_{D_0} = 0.8$ V.

1. **Deduce** the peak output voltage, V_p, and the peak-to-peak ripple voltage, $v_{r(p-p)}$. [2]
2. **Calculate** the average (DC) value of the output voltage. [1]
3. **Estimate** the value of the capacitor from the given data. [2]

Now the input voltage v_i is changed to the one shown in Figure (ii) of the original handout, and the **capacitor is removed**.

4. **Show** the output waveform and **indicate** the voltage levels properly. [2]

Problem 21.4

A voltage waveform $v_i = 8\sin(2000\pi t)$ V is input to a full-wave rectifier. A resistance of $R = 50$ kΩ is connected at the load. Assume that the diodes used in the circuit have a forward drop of 0.8 V.

1. **Draw** the circuit of the full-wave rectifier. Label the input and output voltages properly. [1]
2. **Draw** the waveforms of the input and output voltages. What are the peak values of input and output? Show them in the graph. [1+1]

3. **Find** the average voltage measured at the output. [1]